

An Empirical Evaluation of LLMs for Microservice Bad Smell Detection

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Agenda

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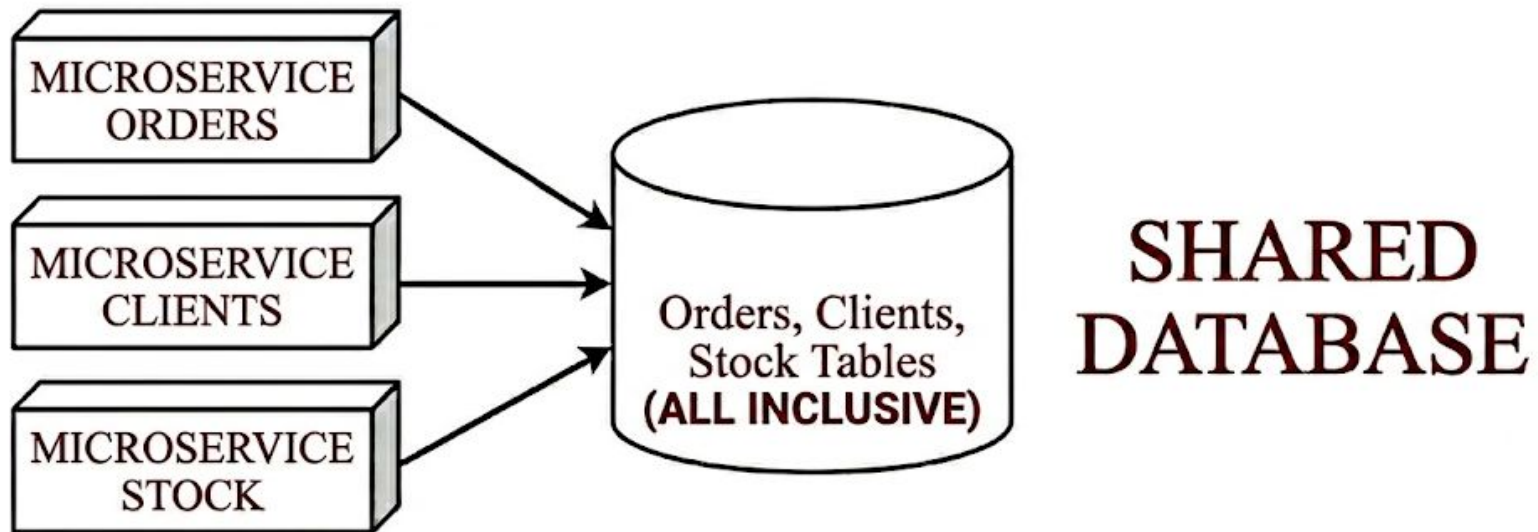
Introduction

- ❑ Microservices are widely adopted by the industry
- ❑ Systems degrades over time due to poor design decisions
- ❑ Detection is:
 - Costly
 - Limited
- ❑ Opportunity: use LLMs 

Background

- Microservices Bad Smells (MBS) are symptoms of poor design
- Examples:
 - No API Gateway
 - Shared Database
 - Cyclic Dependency
 - Shared Libraries

Background



PROBLEM: Various services access and directly modify the same database.

CONSEQUENCE: Tight coupling, scalability difficulties, schema dependency.

Background

- Previous study:
 - Systematic Literature Review on MBS
 - 104 smells
 - 7 detection tools
 - 15 refactoring strategies for the most discussed ones

Related Work

- Traditional MBS detection tools
 - **Static Analysis Limitations**
 - Existing tools often rely on rigid, rule-based heuristics and predefined thresholds to identify smells.
 - **Source Code Dependency**
 - Most traditional approaches require access to source code or execution logs, making high-level architectural analysis difficult without implementation details.

Related Work

- LLMs for Software Architecture
 - **Design Reasoning**
 - Recent studies explore the capability of Large Language Models to reason about architectural trade-offs and quality attributes
 - **Beyond Code Generation**
 - Research is shifting from simple code completion to using LLMs for design pattern recognition and architectural consistency checks.

Related Work

- The research gap
 - **Missing Empirical Evidence**
 - To the best of our knowledge, there are no existing studies specifically investigating the detection of MBS using LLMs.
 - **Architectural Contextualization**
 - While traditional tools excel at structural detection, we explore if LLMs can leverage **architectural context** to identify **subjective or semantic-rich smells** (e.g., Nano Service) that lack clear-cut, rule-based definitions.

Research Questions

- **RQ1:** How effective are LLMs at detecting MBS?
- **RQ2:** How does detection performance vary across different models?
- **RQ3:** How does detection performance vary across different MBS?
- **RQ4:** How does the type of smell (structural vs semantic) affect LLM detection performance?

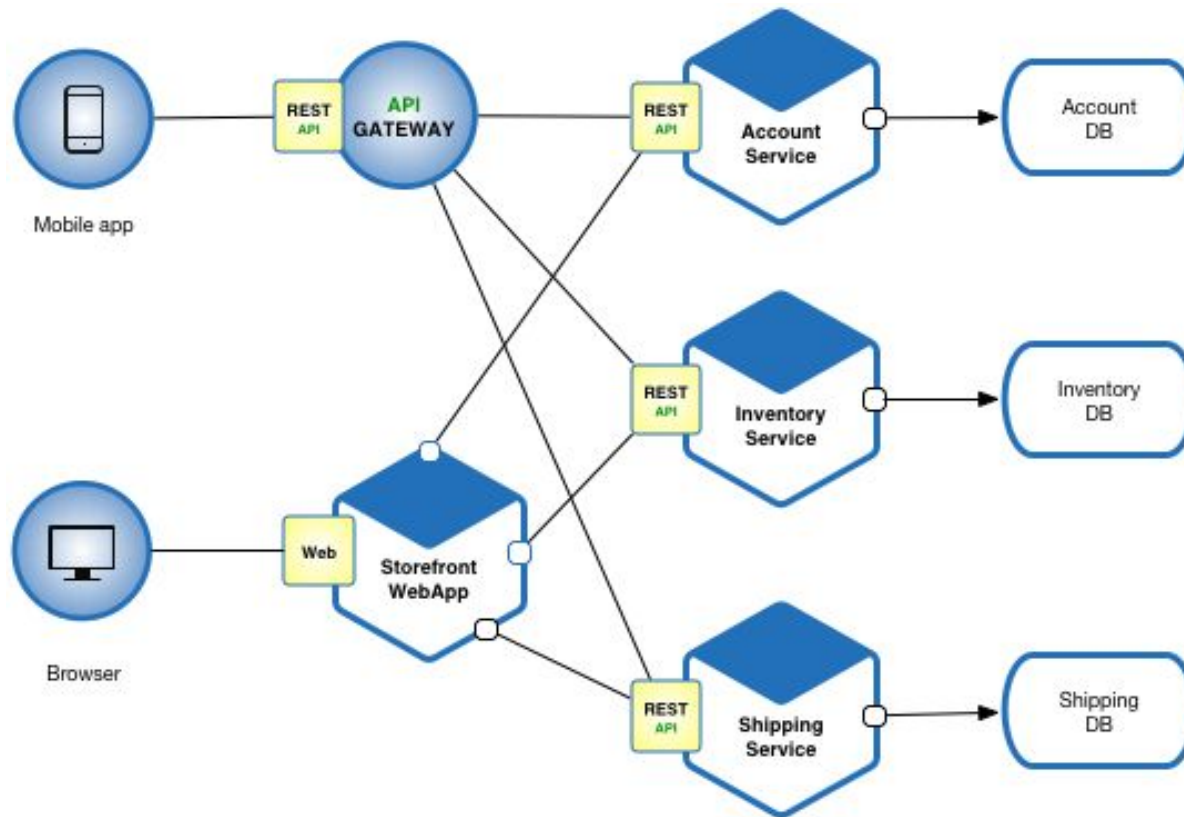
Study Design

- Smell selection
 - Focus on specific smells
 - No API Gateway
 - Nano Service
 - Shared Database
 - Microservice Greedy
 - Cyclic Dependency

Study Design

- Models selection
 - Coding-specific models
 - Claude Opus 4.6
 - GPT-5.4
 - Deepseek-coder-v2
 - Qwen-coder-plus
- Dataset
 - Taibi et al. (2019) microservices dataset
 - Filtered available architecture diagrams

Study Design



Study Design

- Prompting strategy
 - Clear definitions
 - Input the diagram as an image
 - Structured output format

```
{  
  "smell": "Nano Service",  
  "confidence": number,  
  "involved_services": string,  
  "explanation": string  
}
```

Study Design

You are a ****senior software architect**** analyzing microservice architecture diagrams to detect bad smells.

The smell to detect is ****CYCLIC DEPENDENCY****: two or more services form a circular dependency chain (e.g., $A \rightarrow B \rightarrow C \rightarrow A$), making independent deployment impossible for the involved services and complicating fault isolation.

Considering the attached diagram, what is your confidence on the presence of the Cyclic Dependency smell?

- 1 - Not present
- 2 - Unlikely present
- 3 - Likely present
- 4 - Clearly present

Respond only with a JSON object following this schema:

```
{  
  "smell": "Cyclic Dependency",  
  "confidence": number,  
  "involved_services": [string],  
  "explanation": string  
}
```

← Role

← Smell description

← Confidence

← Output format

Study Design

- Ground truth construction
 - Online questionnaire to validate the presence of selected smells in architectural diagrams
 - Confirmation criteria:
 - Smell confirmed by majority agreement ($\geq 50\%$ of participants)
 - Participants:
 - 6-10 Software engineering researchers
 - Level of agreement will be assessed to evaluate reliability

Pilot study

- Goal: validate the experimental setup
- Setup:
 - Small subset of architectures
 - Multiple runs per model
- Findings
 - The model performed better separating prompts per smell
 - Very small variability across runs
- Improvements
 - Separate prompts

Execution

- ❑ Input: architecture + prompt
- ❑ LLM predicts smells
- ❑ Output parsed into structured format
- ❑ Compare with ground truth
- ❑ Multiple runs per configuration

Planned Analysis

- ❑ Evaluate detection performance using Precision, Recall, and F1-score
- ❑ Compare results across different LLMs
- ❑ Analyze performance per microservice bad smell
- ❑ Compare performance across smell types (structural vs. semantic)
- ❑ Assess prediction variability across multiple runs
- ❑ Conduct qualitative error analysis to identify common failure patterns

Expected Insights

- Effectiveness of LLMs in detecting architectural smells
- Differences in performance across models
- Impact of smell type on detection performance
- Strengths and limitations of LLMs in architectural reasoning
- Common failure patterns in LLM-based smell detection

Threats to Validity

- **Construct validity:** Subjectivity in interpreting microservice bad smells from architectural diagrams
- **Internal validity:** Prompt design and model configuration may influence detection results
- **External validity:** Limited number of diagrams and selected smells may affect generalizability
- **Conclusion validity:** Limited number of runs and participants may impact the robustness of results

Conclusion

- ❑ Detecting microservice bad smells is still limited by traditional approaches
- ❑ LLMs offer a promising alternative for architectural reasoning
- ❑ This study provides a systematic evaluation framework for LLM-based smell detection
- ❑ Our goal is to understand not only performance, but also when and why LLMs succeed or fail

Thank you!
Questions?